

Consumption of animal products, their nutrient components and postmenopausal circulating steroid hormone concentrations.

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BACKGROUND/OBJECTIVES: Little is known about nutritional factors that influence circulating concentrations of steroid hormones, which are consistently associated with risk of breast cancer for postmenopausal women. We aimed to investigate the association between consumption of animal products and the plasma concentrations of steroid hormones and sex hormone-binding globulin (SHBG). **SUBJECTS/METHODS:** Cross-sectional analysis was conducted on plasma from 766 naturally postmenopausal women. We measured plasma concentrations of steroid hormones and SHBG, and estimated dietary intakes using a 121-item food frequency questionnaire. Log-transformed values of hormone concentrations were regressed on quartiles of intake of meat and dairy products among food items, and fats, proteins and cholesterol among nutrient intake. **RESULTS:** Total red and fresh red meat consumption was negatively associated with SHBG levels (P for trend=0.04 and <0.01, respectively). Mean SHBG concentrations were approximately 8% and 13% lower for women in the highest quartile compared with the lowest quartile of total red and fresh red meat consumption, respectively. Positive associations were observed between dairy product consumption and total and free estradiol concentrations (P for trend=0.02 and 0.03, respectively). Mean concentrations of total and free estradiol were 15 and 14% higher for women in the highest quartile of dairy product consumption than for those in the lowest quartile, respectively. No associations were observed with consumption of processed meat, chicken, fish, eggs, cholesterol, fats or protein. **CONCLUSIONS:** Our study suggests that greater consumption of total red and fresh red meat and dairy products might influence circulating concentrations of SHBG and estradiol, respectively. Confirmation and further investigation is required.